

LoRA Fine-Tuning on Qwen2.5 — Project Explanation

Overview

This project is the second entry in my LoRA fine-tuning work, applying the same parameter-efficient fine-tuning technique to a real instruction-tuned chat model, Qwen2.5-0.5B-Instruct, rather than a base model with no chat awareness. The fine-tuning goal is also different from the first project: enforcing a strict, exact output format alongside correct answers, not just teaching new factual content.

What the Notebook Does

Training examples are written as full instruction-tuning blocks, each specifying an instruction to follow an exact `HR_RESPONSE` prefix format, an employee message, and the required structured response, mirroring how instruction-tuned models are genuinely trained rather than using plain question-answer pairs. The model is loaded in half precision and moved explicitly onto GPU, and prompts are formatted using the tokenizer's chat template function rather than passed as raw text, since this is the correct interface for any instruction-tuned model and directly affects output quality.

The LoRA configuration targets four separate attention projection layers, query, key, value, and output, reflecting Qwen's architecture in direct contrast to GPT-2's single fused attention layer used in the earlier `distilgpt2` project, concretely demonstrating that LoRA target module configuration is architecture-specific rather than universal. Training runs for fifteen epochs, fewer than the earlier project, reflecting that an already instruction-tuned model requires less adjustment. A clear before-and-after comparison confirms the model adopts the enforced response format after training, and a second, differently worded question tests whether that format compliance generalises beyond the exact training examples.

Skills Demonstrated

- Fine-tuning a real instruction-tuned chat model with LoRA
- Correct chat-template based prompting for instruction-tuned models
- Architecture-specific LoRA configuration across different model families
- Teaching output-format compliance through fine-tuning, not just factual recall

Understanding format-enforcing fine-tuning on instruction-tuned models is directly relevant to the structured-output requirements of my dissertation's agentic intelligence platform.